Band-1 Hadamard Order 6788 — Miyamoto (Kronecker Seed H_1696 + C_1698); Miyamoto-only NOT in SPK; Gap 2052 Still OPEN

H_212 Paley-I PASS; H_424/H_1696 Kronecker PASS; C_1698 PASS; H_6788 Miyamoto PASS (NOT in SPK); gap 2052 still OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-09-15

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_6724_Miyamoto`; `2026-09-15_Band1_Hadamard_6404_Kronecker`.
Lab. `/workspace/experiments/band1/hadamard_6788_miyamoto/`.
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Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 212, 424, 1696, and 6788 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Kronecker seed chain H_{212} = Paley-I($q = 211$), H_{424} = Kronecker(H_2, H_{212}), H_{1696} = Kronecker(H_4, H_{424}) and symmetric Paley conference C_{1698} over prime $q = 1697 \equiv 1 \pmod 4$. Classical SPK for order **6788**: Paley-I **FAIL**s ($q = 6787 = 11 \cdot 617$); Paley-II **FAIL**s ($q = 3393 = 3^2 \cdot 13 \cdot 29$); Kronecker **FAIL**s ($4 \times 1697$). Therefore order **6788 is a Miyamoto-only bonus (NOT in SPK)**. Seeds **212 / 424 / 1696** enter SPK via classical paths. It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under SPK. `clay_closed = false`. Prior skip $q = 1637 \to n = 6548$ retained; intervening $q = 1693 \to n = 6772$ not likely_buildable; $q = 1697$ next buildable. Prior SPK seeds **840 / 1680 / 6724 / 1668 / 6628 / 1656 / 1620 / 1608 / 6436** retained; Miyamoto-only bonuses **6676/6484/6404/6388/6196/6788/...** excluded.

Abstract

Finite constructions of orders **212**, **424**, **1696**, and **6788**.

- H_{212} = Paley-I ($q = 211$)
- H_{424} = Kronecker(H_2, H_{212})
- H_{1696} = Kronecker(H_4, H_{424}) — Miyamoto seed
- C_{1698} = Paley conference over GF(1697), $q = 1697$
- H_{6788} = Miyamoto(H_{1696}, C_{1698})
- Paley-I / Paley-II / Kronecker for 6788: **SKIP / FAIL**

SPK: 6788 Miyamoto-only bonus (NOT in SPK). Seeds 212 / 424 / 1696 enter SPK. Prior SPK **840 / 1680 / 6724 / 1668 / 6628 / 1656 / 1620 / 1608 / 6436** retained.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

```
python3 verify_hadamard_6788.py
```

Results. overall_pass=true; units 106/106; wall 251.7942s; all max|HH^T-nI|=0.

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Arithmetic

- Target $n = 6788 = 4 \cdot 1697$; 1697 prime $\equiv 1 \pmod 4$ → Miyamoto-eligible.
- Need H_{1696} and C_{1698} (Paley conference over prime $q = 1697$).
- Seed chain: Paley-I $q = 211$ → H_{212} → Kronecker(H_2, ·) → H_{424} → Kronecker(H_4, ·) → H_{1696}.
- SPK: Paley-I $q = 6787$ **FAIL**; Paley-II $q = 3393$ **FAIL**; Kronecker **FAIL** → **Miyamoto-only (NOT in SPK)**.
- Prior skip: $q = 1637 \to n = 6548$. Intervening: $q = 1693 \to n = 6772$ not likely_buildable; $q = 1697$ next buildable.

Constructions (verified)

Object	Method	max|HH^T-nI|	Wall (s)
H_212 Paley-I	Paley-I q=211	0	0.0199
H_424 Kronecker	Kronecker(H_2, H_212)	0	0.0364
H_1696 Kronecker	Kronecker(H_4, H_424)	0	2.2028
C_1698	Paley conf q=1697	0	3.3692
H_6788 Miyamoto	Miyamoto(H_1696 Kronecker, C_1698)	0	244.4255
H_6788 Paley-I	SKIP q=6787=11 · 617 not prime	—	—
H_6788 Paley-II	SKIP q=3393=3^2 · 13 · 29 not prime	—	—
H_6788 Kronecker	SKIP 4 · 1697 not admissible	—	—

SPK closure note

Seeds include prior SPK ships through **6724** lineage (**6724** Paley-II; **840/1680**; **6628**; **1668**; **1656**; **1620**; **1608**; **6436**; **NOT** 6676/6484/6404/6388/6196/6788) + verified **212** / **424** / **1696** (classical); **exclude** Miyamoto-only bonuses including **6788**. Known count 1392. Next constructive gap **2052** (OPEN).

Quantity	Value
6788 in SPK	**False** (Miyamoto-only)
1696 in SPK	**True** (Kronecker)
424 in SPK	**True** (Kronecker)
212 in SPK	**True** (Paley-I q=211)
840 / 1680 / 6724 in SPK	**True** (prior)
1668 / 6628 / 1656 / 1620 / 1608 / 6436 in SPK	**True** (prior)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Non-claims

No full Hadamard conjecture · no Clay-closed · no finished-ToE · no RH / P NP / YM / Frankl / NS-solved / AC-solved · gap 2052 remains OPEN · 6788 is NOT folded into SPK.